Course Code: 23ITE029		Course Title: Machine Learning with Python	
Course Category: Major		Course Level: Higher	
L:T:P(Hours/Week)2: 0: 2	Credits: 3	Total Contact Hours: 60	Max Marks: 100

Course Objectives

This course is intended to cover the fundamentals of machine learning, including data preprocessing, feature engineering, and the implementation and evaluation of both supervised and unsupervised algorithms, and exploring advanced techniques.

Module I **15 Hours**

Introduction - Scikit-learn-Essential Libraries and Tools-Machine Learning Types

Supervised Learning: Generalization, Overfitting, and Underfitting- Linear Models –Naive Bayes Classifiers–Ensembles of Decision Trees- Kernelized Support Vector Machines

Unsupervised Learning: Types of Unsupervised Learning - Preprocessing and Scaling–Dimensionality Reduction: Principal Component Analysis (PCA)-Manifold Learning with t-SNE

Module II **15 Hours**

Data Representation and Feature Engineering: Categorical Variables- Binning, Discretization, Linear Models, and Trees-Interactions and Polynomials- Automatic Feature Selection

Model Evaluation and Improvement: Cross Validation- Grid Search- Evaluation Metrics and Scoring: Metrics for Classification- Regression Metrics- Algorithm Chains and Pipelines

Recent advancement: Introduction to Transformers and its architecture- Text Processing Transformers-Vision Transformer

List of Experiments: **30 hours**

1. Work with NumPy, Panda by Loading Dataset in Python and Implement visualization with Matplotlib and Seaborn.
2. Implement Support Vector Machine (SVM) for Classification on a real-world dataset

3. Implement Ensemble Learning Techniques: Bagging and Boosting
4. Implement Dimensionality Reduction using PCA and t-SNE
5. Implement image classification using Pretrained Vision Transformers
6. Develop a mini project focused on predictive modeling.

Course Outcomes	**Cognitive Level**
At the end of this course, students will be able to:	
CO1: Apply the fundamentals of supervised and unsupervised learning to implement key algorithms and solve real-world problems.	Apply
CO2: Implement machine learning models like linear models, decision trees, and support vector machines using scikit-learn.	Apply
CO3: Apply dimensionality reduction techniques like PCA and t-SNE for data preprocessing and feature extraction.	Apply
CO4: Analyze the model performance through techniques like feature engineering and model optimization.	Analyze

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	2	-	-	-	-	-	-	-	-	-		-
CO3	-	-	-	2	-	-	-	-	2	-	-	-	-	2
CO4	-	3	-	-	3	-	-	-	-	-	2	-	3	-

High-3; Medium-2; Low-1

Text Book(s):

T1. Andreas C. Muller, Sarah Guido, “Introduction to Machine Learning with Python”, 4th Release,O’Reilly,2018.

T2. Uday Kamath, Kenneth Graham, Wael Emara, “Transformers for Machine Learning: A Deep Dive” ,1st Edition, CRC Press, 2022

Reference Book(s):

R1. Dipanjan Sarkar, Raghav Bali, Tushar Sharma, “Practical Machine Learning with Python”, 1st Edition,Apress, 2018

R2. John D. Kelleher ,Brian Mac Namee , Aoife D'Arcy, “Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies”, 2nd Edition,The MIT Press,2020.

R3. Oliver Theobald, “Machine Learning for Absolute Beginners”, 3rd Edition, Scatterplot Press, 2017.

R4. Ethem Alpaydin, “Introduction to Machine Learning”, 4th Edition, The MIT Press, 2020.

Web References:

1. https://scikit-learn.org/stable/
2. https://www.tutorialspoint.com/machine_learning_with_python/index.html
3. https://www.w3schools.com/python/python_ml_getting_started.asp